MIL-B-13164A(ME)
14 December 1973
SUPERSEDING
MIL-B-13164(CE)
11 December 1953

MILITARY SPECIFICATION

BRIDGE, FIXED, STEEL, HIGHWAY,

SEMIPERMANENT, 90-FOOT

TRUSS SPAN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers components for a class 50, fixed, semi-permanent, through truss, highway bridge of 90-foot span length.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-575	- Bolts, Hexagon and Square.
FF-B-584	- Bolts, Finned Neck; Key Head; Machine; Ribbed Neck; Square Neck; Tee Head.
FF-N-105	- Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-W-84	- Washer, Lock (Spring).
FF-W-92	- Washers, Metal, Flat (Plain).

FSC 5420

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QQ-I-652	- Iron Casting, Gray.
QQ-S-634	- Steel, Bar, Carbon, Cold Finished, (Standard Quality).
QQ-S-698	- Steel, Sheet and Strip, Low Carbon.
QQ-S-781	- Strapping, Steel, Flat and Seals.
QQ-S-790	- Steel Strapping, Round (Bare and Zinc-Coated).
TT-W-571	- Wood Preservation: Treating Practices.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-R-1223	- Rivets and Rivet Steel Bars.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

D 2658-1 to -4 inclusive - Steel Highway Bridge, 90 Foot Truss Span.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel
A123 - Zinc (Hot-Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Bars, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
B3.0 - Standard Qualification Procedure.
D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

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WESTERN WOOD PRODUCTS ASSOCIATION

Standard Grading Rules

(Application for copies should be addressed to the American Wood Products Association, 700 Yeon Building, Portland, OR 97204.)

SOUTHERN PINE INSPECTION BUREAU

Standard Grading Rules

(Application for copies should be addressed to the Southern Pine Inspection Bureau, P.O. Box 52468, New Orleans, LA 70150.)

3. REQUIREMENTS

3.1 Description. The bridge component(s), as applicable, shall be as shown on Drawings D 2658-1 to -4 inclusive, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a bridge component(s), as applicable, for inspection within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods will produce bridge components that comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

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3.3.1 Structural steel. Structural steel shall conform to ASTM A36.

3.3.2 Steel rods and bars. Steel rods and bars shall conform to QQ-S-634, designation 1018 or 1020.

3.3.3 Lumber. Lumber shall be Douglas fir (1200f, 1000c) conforming to the Standard Grading Rules for Western Lumber; or Dense No. 1 Structural grade conforming to the Standard Grading Rules for Souther Pine Lumber. Moisture content shall be not more than 19 percent. Lumber shall be roughed out to the sizes shown on the drawings.

3.3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1, classification and size as required.

3.3.5 Rivets. Rivets shall conform to MIL-R-1223, type B-2 or type C-1, as specified on drawings.

3.4 Floor clips. Floor clips shall be of steel conforming to QQ-S-698, hot rolled, as-rolled condition, and shall be hot-galvanized, after fabrication, in accordance with ASTM A123.

3.5 Bearing spools. Bearing spools shall be iron castings conforming to QQ-I-652, class 20. Tolerances for finished dimensions shall be plus or minus 1/32 inch.

3.6 Bolts, nuts, and washers.

3.6.1 Bolts. Unless specified herein, bolts shall conform to FF-B-575, type I and type II, carbon and alloy steel.

3.6.1.1 Structural bolts. Structural bolts shall conform to FF-B-584, type III, made from medium carbon having a minimum tensile strength of 70,000 psi. Bolts shall provide a full section in shear.

3.6.1.2 Anchor bolts. Anchor bolts shall be of steel conforming to 3.3.2.

3.6.2 Nuts. Nuts shall conform to FF-N-836, type I, style 1, and type II, style 4. The nuts shall be made from grade C steel.

3.6.2.1 Structural bolt nuts. The structural bolt nuts shall be semi-finished, recessed, and hexagonal shaped.

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3.6.3 Threaded components. Threads for bolts and nuts shall be Unified thread series, classes 2A and 2B. Structural bolts shall be threaded with self-locking type threads.

3.6.4 Washers. Washers for 3/4-inch bolts shall conform to FF-W-92, type A, grade I. "OG" washers shall be of cast iron, 4-1/8-inch diameter, 1-inch thickness, 1-1/8-inch diameter hole. Lock washers shall conform to FF-W-84, class A, style 3 for 1-inch spring-lock washers.

3.7 Nails. Nails shall conform to FF-N-105, common nail, cement-coated, 50 d, steel.

3.8 Identification marking. The components shall be identified in accordance with MIL-STD-130.

3.9 Treatment and painting. All components shall be treated and painted in accordance with MIL-T-704, type A. Component items which are furnished in accordance with separate specifications shall be treated and painted in accordance with the end item specification. In lieu of painting, all connecting bolts, nuts, and threaded portions of components shall be preserved in accordance with MIL-P-116, method I, type P-1. Lumber shall not be painted, but shall be treated in accordance with TT-W-571, creosote-petroleum solution, after all cutting has been completed.

3.10 Stenciling. The nomenclature, gross weight, and erection mark of each major component of the bridge shall be stenciled on each side of the component near the center. Nomenclature shall be 3-inch, Gothic-type capitals; weight shall be in symbols and Arabic numerals 2 inches high. Markings shall be 2 inches high when the size of the component will not permit the use of 3-inch characters. Stenciling shall conform to MIL-T-704.

3.11 Workmanship. All parts, components, and assemblies of the bridge shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material.

3.11.1 Fabrication. Steel used in the fabrication of the components shall be free from laminations, kinks, and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Where re-entrant cuts cannot be made by shearing, a rectangular punch or router may be used. All burrs, rough edges, and sharp edges of holes shall be ground smooth. Sawing, shearing, and chipping of steel used infabrication shall be done neatly and accurately. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks.

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3.11.2 Rivet and boltholes. All bolts for field connections, holes in gusset plates, and holes in members through which gusset plates are shop-riveted shall be accurately drilled through steel templates after all welding and riveting have been completed or shall be subpunched prior to welding and riveting and then reamed to full size through a steel template after welding and riveting. Holes for shop-driven rivets and ribbed bolts may be punched full size provided the holes are within the tolerances specified for position. The diameter of the die shall not exceed the diameter of the punch by more than 1/16 inch. Where holes are subpunched and reamed, the subpunched holes shall be 1/4 inch less in diameter than the hole size shown on the drawings. Where countersunk holes are required, the countersinking shall be uniform and concentric with the holes and of sufficient depth to permit an adequate rivet head. All holes shall be smooth and cleancut without torn or ragged edges.

3.11.3 Shop riveting. Parts to be shop riveted shall be tightly bolted in position before riveting. Rivets shall be driven hot with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of high button (acorn) shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, and in full contact with the surfaces of the member. All countersunk rivets shall be chipped or ground flush.

3.11.4 Welders and welding.

3.11.4.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

3.11.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1-Structural

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Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.11.5 Tolerances. The position of all holes for field connections shall be within plus or minus 1/32 inch of their true position. The diameter of all drilled holes shall be within plus or minus 1/64 inch of the diameter shown on the drawings. The tolerance for overall dimensions of members shall be within plus or minus 1/16 inch of those shown on the drawings. The tolerances specified for the positioning of holes and material and the length, width, and depth of member shall be referenced from one end or side of the member and shall be noncumulative.

3.11.6 Interchangeability. Like components shall be fabricated to be duplicates within the tolerances specified.

3.11.7 Machine work. Finished contact and bearing surfaces marked by an "f" on the drawings shall be machined true and exact. Bearing and sole plates shall be machined from plates not less than 1/16 inch thicker than the finished thickness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

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4.3 Preproduction inspection. The bridge component(s) as applicable shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The bridge component shall be examined as specified herein, for the following defects:

Major

101. Material not as specified.
102. Components missing or not as specified.
103. Workmanship not as specified.
104. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
105. Members not true to dimension and line, or containing twists or bends.
106. Rivets missing, misshapen, or incorrectly driven.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.
203. Stenciling not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.6.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.
108.	Unprotected surfaces not coated with preservation as specified for level A.
109.	Bolts, nuts, and washers not packaged as specified for level A.
110.	Nails not packaged and packed in accordance with the referenced document.
111.	Small components not packed as specified for level A.
112.	Large bundled components not secured as specified for level A.
113.	Marking missing, illegible, incorrect, or incomplete for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unpainted metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Bolts, nuts, and washers. Bolts, nuts, and washers of like description shall be packaged together in close-fitting boxes conforming to PPP-B-636, W5c or W6c, style optional, in quantities not to exceed the weight limitations of the box. The boxes shall be closed and sealed as specified for method V in the appendix to the box specification.

5.1.1.3 Nails. Nails shall be packaged in accordance with the level A requirements of FF-N-105.

5.1.2 Level C. The bridge components shall be preserved and packaged in a manner to afford protection against deterioration and damage from the supplier to the initial destination.

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5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Bolts, nuts, and washers. Bolts, nuts, and washers, preserved and packaged as specified in 5.1, shall be packed together in quantities not to exceed 200 pounds gross weight in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. The boxes shall be closed and strapped in accordance with the appendix to the box specification.

5.2.1.2 Nails. Nails, packaged as specified in 5.1, shall be packed in accordance with the level A requirements of FF-N-105.

5.2.1.3 Small components. Small components shall be packed in boxes or secured in bundles in the most convenient manner that will prevent loss or damage to the components and provide aid in segregation, erection, and ease of handling. Bundled components shall be limited to a gross weight not to exceed 200 pounds. Boxed components shall not exceed the weight limitation of the box. To the extent practical only components of like description shall be contained within the same bundle or box. Bundled components shall be secured with steel strapping conforming to QQ-S-790, class B, finish 2; or QQ-S-781, finish B. Boxes shall conform to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, grade B, style optional. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall be zinc-coated.

5.2.1.4 Large components. Large components shall be shipped loose or in secured bundles. Bundles shall not contain components of more than one size or description and shall be secured with strapping conforming to QQ-S-781, type I or IV, finish B, size 1-1/4 by 0.035 inches. A minimum of two straps shall be used for lengths up to 6 feet, three straps for lengths up to 12 feet, and one additional strap for each additional 6 feet in length. Bundles shall be confined to a minimum of 6,000 pounds gross weight.

5.2.2 Level C. The bridge components shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages, bundles, and loose components shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The 90-foot steel highway bridge is a semipermanent structure intended primarily to replace tactical (or combat) bridges in lines of communication.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production bridge components from the approved preproduction model(s), as applicable, during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish components conforming to this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 5420-A249

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-13164A(ME)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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